Dr Yashodhan P. Gokhale, with over 18 years of experience, is the Vice President for Battery Technology at JSW Energy. Before joining JSW Energy, Dr Gokhale worked in several leadership positions, including CTO for Battrixx, Kalyani Powertrain Ltd, Octillion Power Systems, and Tata Motors Research Centre, Bengaluru. He leads the advancement of battery technologies, including lithium-ion, solid-state, and sodium-ion. Dr Gokhale, a PhD graduate in nanoscience from Otto Von Guericke University, Germany, has been recognised by numerous publications and awards in energy storage.

Dr Gokhale's expertise is not just theoretical but also highly practical. His proven track record in developing nano-based and gas diffusion electrodes and optimising battery thermal management systems has real-world applications. His adept management of sales and negotiations, along with his ability to secure orders for energy storage devices across India, demonstrates the practical impact of his work.

handled properly. Safety largely depends on the quality of the battery management system (BMS), which is crucial for preventing issues like thermal propagation that can lead to fires. In India, regulatory measures like AIS 156 for materials traceability and AIS 038 for vehicles have been introduced to enhance safety by ensuring that thermal propagation in batteries is contained.

Are the components used in low-voltage and high-voltage EV batteries the same, or are there significant differences?

There are significant differences, particularly in the wiring harnesses, fuses, and connectors, which must handle higher voltages, such as 350, 650, or 800 volts. Low-voltage components cannot be used for high-voltage applications due to their lower current-carrying capacity, which makes them susceptible to damage during operation.

Considering that India primarily uses low-voltage batteries, especially in two- and three-wheelers, do you see the shift to high-voltage batteries as an opportunity for MSMEs in the component ecosystem?

Absolutely, this transition offers a substantial opportunity for MSMEs. The current low-voltage battery market in India only sees about 10-20% value addition locally, which is a significant loss. However, as India shifts towards high-voltage batteries, MSMEs can contribute to manufacturing critical components like wiring harnesses, flexible bus bars, and thermal management systems like cold plates. This shift is not just about assembling imported components but increasing the 'Make in India' percentage, which the government mandates to be over 50%. This approach supports local manufacturing and aligns with the needs of diverse Indian climates, from the cold of Jammu and Kashmir to the humidity of Guwahati and the heat of Chennai and Nagpur. The upcoming PLI schemes and the focus on local production could be game changers in India's EV ecosystem.

How do battery OEMs typically source their components, and what criteria determine whether they partner with MSMEs or startups?

OEMs generally follow a standardised process, such as PPAP (production part approval process), when selecting suppliers. Many components, especially for batteries, were initially sourced from countries like China, Korea, Japan, or Europe due to quality and cost considerations. However, with the push for self-reliance ('*Atmanirbhar Bharat*') and increased awareness of the risks of low-quality imports—like thermal events due to poor components—there is a shift towards sourcing